SCHOOL OF ARCHITECTURE, COMPUTING & ENGINEERING

INDIVIDUAL CW A REPORT

Module Leader: Dr. Elias Eze

Assignment tutor-Dr. Aloysius Edoh PhD and Dr. Athirah Mohd Ramly

Student Name: SORABH NEGI

Student Number: u2824224

Module Code: CN7029

Term: 1

TABLE OF CONTENTS

INTRODUCTION.....3

OBJECTIVES.....3

DESIGN AND BUILD COMPUTER NETWORK USING PACKET TRACER.....3

TASK 1 IP CALCULATION.....4

TASK 2 ACTIVITY OF COURSEWORK.....4

TASK 3 OVERVIEW OF SUBNETS AND THE HOST DEVICES 4

TASK 4 CONFIGURE THE PHYSICAL TOPOLOGY.....6

TASK 5 CONFIGURE THE LOGICAL TOPOLOGY.....6

SIMULATION.....10

TASK 6 PING AND PDU.....26

NETWORK DESIGN.....21

RESULT AND ANALYSIS (NETWORK VERIFICATION) 21

EVALUATION.....31

CHALLANGES.....31

CONCLUSION.....32

REFERENCES.....33

TABLE OF TABLES

TABLE 1 Subnets.....5

TABLE 2 Overview of Subnets and host devices.....5

TABLE 3 Design.....6

TABLE 4 Configuring the physical topology 7

TABLE 5 Network C (wire Subnet) 7

TABLE 5.1.....8

TABLE 5.2.....8

TABLE 5.3.....8

TABLE 5.4.....8

TABLE 6 Network B (wired subnet) 9

TABLE 6.1.....9

TABLE 6.2.....10

TABLE 6.3.....10

TABLE 7 Network A.....10

TABLE 7.1.....11

TABLE 7.2.....11

TABLE 7.3.....11

TABLE 7.4.....11

TABLE 7.5.....11

TABLE 7.6 12

TABLE 8 SUBNET D (PING AND PDU) 25

TABLE 8.1 SUBNET F (PING AND PDU) 26

TABLE 8.2 SUBNET E (PING AND PDU) 27

TABLE 9 PINGING DEVICES SUBNET C TO OTHER SUBNETS 28

TABLE 9.1 PINGING DEVICES SUBNET B TO OTHER SUBNETS 28

TABLE 9.2. PINGING DEVICES SUBNET A TO OTHER SUBNETS 29

TABLES OF FIGURES

FIGURE 1 ROUTER 0 CLI.....13

FIGURE 2 ROUTER 0 CLI..... 14

FIGURE 3 ROUTER 0 CLI.....14

FIGURE 4 ROUTER 1 CLI.....15

FIGURE 5 ROUTER 1 CLI.....16

FIGURE 6 ROUTER 1 CLI.....17

FIGURE 7 ROUTER 2 CLI.....18

FIGURE 8 ROUTER 2 CLI.....18

FIGURE 9 ROUTER 2 CLI.....19

FIGURE 10 ROUTER 2 CLI.....20

FIGURE 11WIRELESS ROUTER GUI 20

FIGURE 12 PC0 STATIC CONGIFURED IP.....21

FIGURE 13 SERVER 0 STATIC CONFIGURED IP 23

FIGURE 14 SERVER 1 STATIC IP 23

FIGURE 15 LAPTOP IP IN SUNET A.....24

FIGURE 16NETWORK DESIGN.....25

FIGURE 17 COMPLEX PDU FROM ROUTER 0..... 26

FIGURE 18 COMPLEX PDU FROM ROUTER 1..... 26

FIGURE 19 COMPLEX PDU FROM ROUTER 2.....29

FIGURE 20 PDU LIST.....31

FIGURE 21 PING FROM SUBNET A..... 31

FIGURE 22 PING FROM SUBNET A.....32

FIGURE 23 PING FROM SUBNET B.....32

FIGURE 24 PING FROM SUBNET B.....33

FIGURE 25 PING FROM SUBNET A.....33

FIGURE 26 PING FROM SUBNET A.....33

INTRODUCTION

This report document is the explanation of designed (topology), implementation and simulation of a hybrid wired and wireless network using Cisco Packet Tracer. Calculations of IP addresses according to student ID 2824224. Subnetting i.e., classless, network masks, routing protocols (RIP v2) and wireless security (SSID, WPA2-PSK). The network contains wired subnets for servers and desktop and a wireless subnet for mobile devices, illustration security and routing configuration. The ipv4 is designed by VLSM (variable length subnet masks), wireless router relates to router which is using DHCP (Dynamic host control protocol) pool for the Ip distribution.

OBJECTIVES

The objective of this coursework is to successfully well-built secured network system using Cisco Packet Tracer for the illustration of working devices.

- This involves the personalised IP address using the provided instructions in the coursework
- Use of VLSM to get the other subnets to be used by the devices.
- The vital part is creation of a logical topology which is ring topology of routers in subnet D, E and F where they are configure with RIPv2
- Using routers, switches, servers, and other devices. The address should be well configured to be working properly.
- For wireless security, WPA2-PSK is used to provide confidentiality as well as safeguard the access.
- The network is checked after the configuration and surely it is successful in networking with one another. Conclusively, the network is set up with all the configuration and settings to ensure that it will run efficiently.

DESIGN AND BUILD COMPUTER NETWORK USING PACKET TRACER (JUSTIFY THE HYBRID TOPOLOGY AND EXPLAIN WIRELESS SECURITY CHOICES).

Topology

This network is designed as Ring topology Three routers (router0.router1 and router2) connected to one another via a serial DCE; router1 serves as subnet A; router0 serves as subnet B; and router2 as subnet C. Each in router0 subnet is connected to a local switch to connects pc and server. Subnet sizes chosen are /29 to supply host counts economically. A/30 for the router-to-router link. The wired subnets are used for highly reliability to each other while wireless is better for mobility access.

IP addressing: Personal block derived from the student ID. Router 2 is connected to wireless router then to laptops where router is password protected.

Task 1

TASK 1

The IP is calculated by the student number as per the given algorithm in the coursework part 1 to determine block IP address. Likewise, my student my id is **u2824224.**

- In my student ID The first three digits is more than 223 like mine is 282, then I take 191 as the first section of IP address.
- Taking the next part 2 digits (digit 4th and 5th of your student number) for the second section the IP address.
- The next 2 digits (digit 4th and 5th of your student number) for the second section the of IP address is my Ip is 191.42.24.0/25.

Task the first table below shows the 6 subnets to be configured in this project. 3 routers will be needed to allow for communication between Subnets D, E and F.

Table 1

subnets	Number of Hosts
Subnet A	1 access point and 3 laptops
Subnet B	1 server and 3 computers
Subnet C	1 server and 4 computers
Subnet D	None
Subnet E	None
Subnet F	None

Task 2. Activity must be done in the coursework.

The Cisco packet tracer user and the information provided and creating a network. This requires all the networking devices in the network can communicate with each other.

TASK 3– ANALYSIS AND DESIGN OF LOGICAL NETWORK

Overview of Subnets and the host devices

Table 2

subnets	Network Address	Mask	First Host Address	Last Host Address	Broadcast	Bit Mask
Subnet C	191.42.24.0	255.255.255.248	191.42.24.1	191.42.24.6	191.42.24.7	/29

Subnet B	191.42.24.8	255.255.255.248	191.42.24.9	191.42.24.14	191.42.24.14	/29
Subnet A	191.42.24.	255.255.255.248	191.42.24.17	191.42.24.22	191.42.24.23	/29
Subnet D	191.42.24.24	255.255.255.248	191.42.24.25	191.42.24.26	191.42.24.27	/30
Subnet E	191.42.24.28	255.255.255.248	191.42.24.29	191.42.24.30	191.42.24.31	/30
Subnet F	191.42.24.32	255.255.255.248	191.42.24.33	191.42.24.34	191.42.24.35	/30

DESIGN

Table 3 With the assistance of Table 1 the subnetting will be done, with the number of host bits reserved for the host ID for each subnet. The prefix will be taken from subtracting the host bits required from the total 32 bits in Ip address.

Table 3

Subnets	Number of hosts	Nearest Decimal	Host Bits Needed	Prefix	Subnetting Order
Subnet A	1 access point and 3 laptops	8	3	/29	3
Subnet B	1 server and 3 computers	8	3	/29	2
Subnet C	1 server and 4 computers	8	3	/29	1
Subnet D	None	4	2	/30	4
Subnet E	None	4	2	/30	5
Subnet F	None	4	2	/30	6

VLSM subnetting

Since the 2^n-2>= Number of devices in network.

For subnet A, B and C each had 6 devices:

So, n = 3 host bits needed so 2^3-2= 6

For subnet D, E and F each had 2 devices: 2^2 –2=2

So, n=2 host bits needed.

Nearest decimal (Block size) 2^n=4

Prefix length Calculation= 32- n; for n= 3: 32-3 =/29, For n=2: 32-2= /30

Task3.

Calculating VLSM explanations of bit masks

3 host bits = 1111 1111. 1111 1111. 1111 1111. 1111 1000 = 255.255.255.248 subnet mask = /29 (bit mask)

2 host bits = 1111 1111. 1111 1111. 1111 1111. 1111 1100 = 255.255.255.252 subnet mask= /30 (bit mask)

Task 4 CONFIGURING THE PHYSICAL TOPOLOGY

Table 4

Connection Point	Type of Cable / Medium Used
Between Routers and Switches	Copper Straight-through UTP
Between Routers	Serial DTE
Between Router and Wireless Access Points	Copper Straight-through UTP
Between Switches and Computers	Copper Straight-through UTP
Between Switches and Servers	Copper Straight-through UTP
Between Wireless Devices and Access Point	wireless

TASK 5 CONFIGURE THE LOGICAL TOPOLOGY

TASK 5.1(a): **Network C (wired subnet)**

Table 5

(server2)	
IP Address	191.42.24.2
IP Mask	255.255.255.248
Gateway Address	191.42.24.1

Table 5.1

Host pc0	
IP Address	191.42.24.3
IP Mask	255.255.255.248
Gateway Address	191.42.24.1

Table 5.2

Host pc1	
IP Address	191.42.24.4
IP Mask	255.255.255.248
Gateway Address	191.42.24.1

Table 5.3

Host pc2	
IP Address	191.42.24.5
IP Mask	255.255.255.248
Gateway Address	191.42.24.1

Table 5.4

Host5 pc3	
IP Address	191.42.24.6
IP Mask	255.255.255.248
Gateway Address	191.42.24.9

For network B (wired subnet)

Table 6

(server 0)	
IP Address	191.42.24.10
IP Mask	255.255.255.248
Gateway Address	191.42.24.9

Table 6.1

Host pc4	
IP Address	191.42.24.11
IP Mask	255.255.255.248
Gateway Address	191.42.24.9

Table 6.2

Host pc5	
IP Address	191.42.24.12
IP Mask	255.255.255.248
Gateway Address	191.42.24.9

Table 6.3

Host pc5	
IP Address	191.42.24.13
IP Mask	255.255.255.248
Gateway Address	191.42.24.9

FOR NETWORK A (wireless subnet) In this subnet router 2 is configured with DHCP pool to assign the IP address. Wireless Router is secured with WPA2-psk with AES encryption. Also, for the better speed (strong specification) and security purposes.

Table 7

Device	Ip Address	IP Mask	Gateway Address	SSID	Security/passkey
Wireless Access point	191.42.24.18	255.255.255.248	191.42.24.17	networking	sorabhnegi
Laptop 0	191.42.24.21	255.255.255.248	191.42.24.17	networking	sorabhnegi
Laptop 1	191.42.24.19	255.255.255.248	191.42.24.171	networking	sorabhnegi
Laptop 2	191.42.24.20	255.255.255.248	191.42.24.7	networking	sorabhnegi

Subnet D: -

Table 7.1

Router 0 serial 0/0/0	
IP address	191.42.24.25
Ip mask	255.255.255.252
Gateway address	/

Rip	191.42.0.0

Table 7.2

Router 0 serial 0/1/0	
IP address	191.42.24.29
Ip mask	255.255.255.252
Gateway address	
Rip	191.42.0.0

Table 7.3

Router 1 0/0/0	
IP address	191.42.24.33
Ip mask	255.255.255.252
Gateway address	/
Rip	191.42.0.0

Table 7.4

Router 1 serial 0/1/0	
IP address	191.42.24.26
Ip mask	255.255.255.252
Gateway address	/
Rip	191.42.0.0

Table 7.5

Router 2 serial 0/0/0	
IP address	191.42.24.34
Ip mask	255.255.255.252
Gateway address	/
Rip	191.42.0.0

Table 7.6

Router 2 serial 0/1/0	
IP address	191.42.24.30
Ip mask	255.255.255.252

Gateway address	/
Rip	191.42.0.0

SIMULATION

ROUTER 0

Router 0 for this router first there is enable password which is negi. The I configured Fast Ethernet 0/0 Ip 191.42.24.9 subnet 255.255.255.248. For configuring router rip there Ip is 191.42.0.0 and the serial0/0/0 Ip address 191.42.24.25 subnet 255.255.255.252 and serial 0/1/0 191.42.24.29 subnet 255.255.255.252 is configured with subnet D.

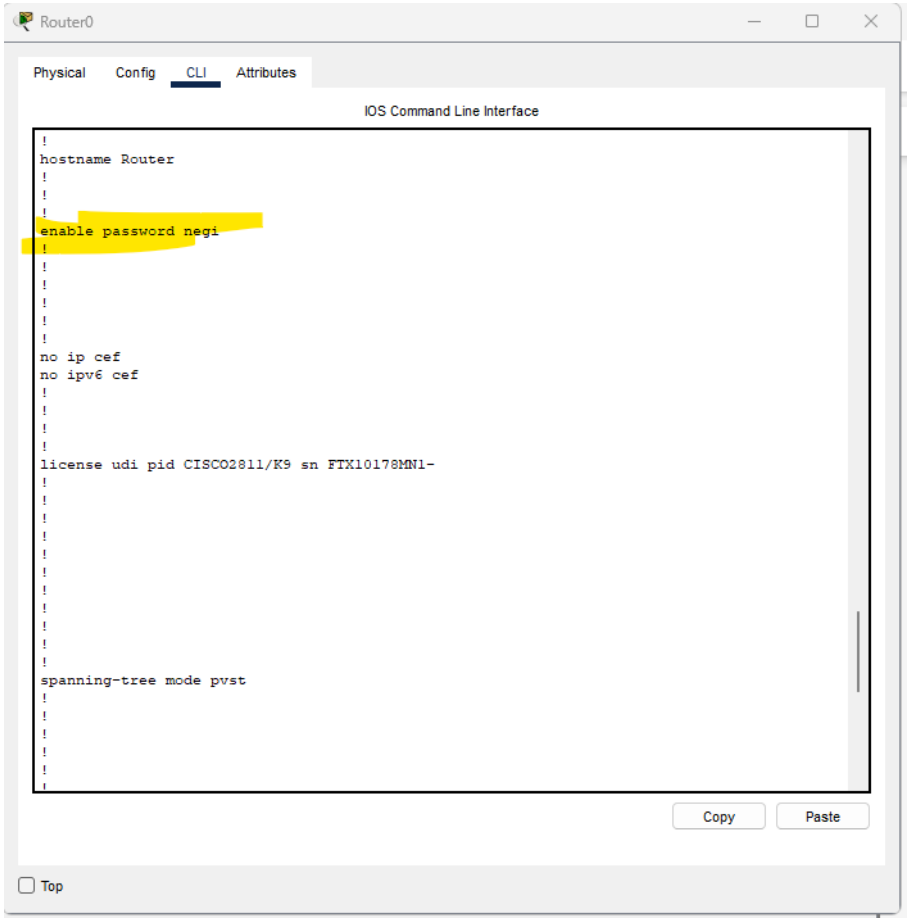


Figure1

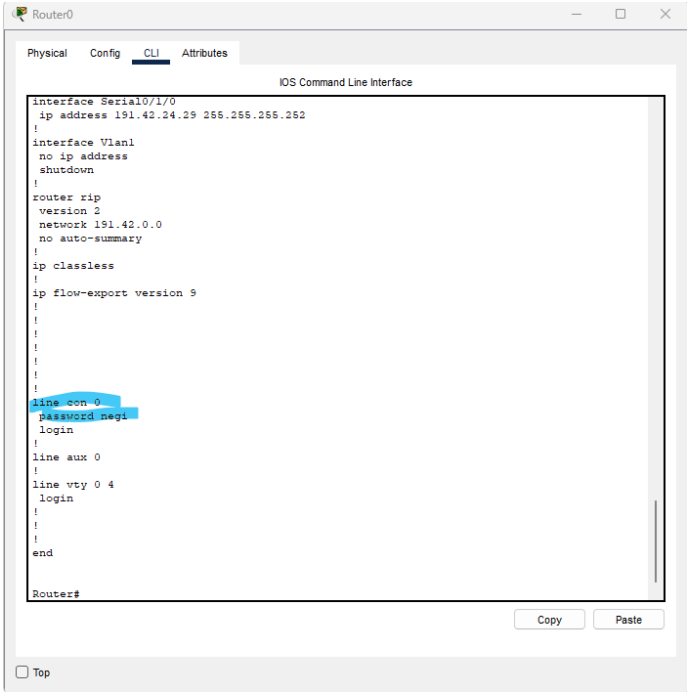


Figure 2

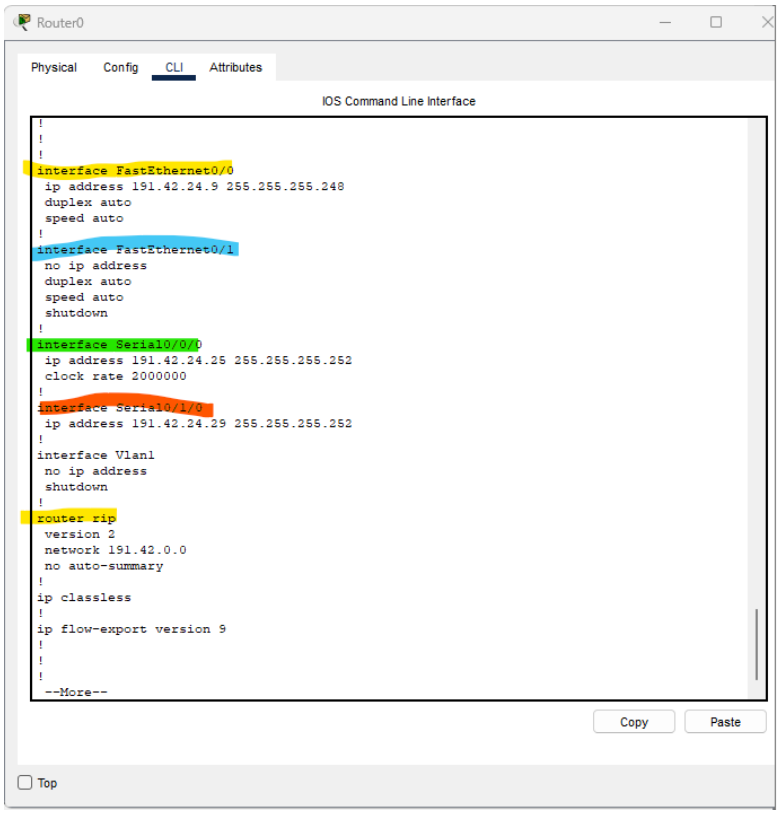


Figure 3

ROUTER 1

Router 1 for this router first there is enable password which is negi. The I configured Fast Ethernet 0/0 Ip 191.42.24.9 subnet 255.255.255.248. For configuring router rip there Ip is 191.42.0.0 and the serial0/0/0 Ip address 191.42.24.26 subnet 255.255.255.252 and serial 0/1/0 191.42.24.33 subnet 255.255.255.252 is configured with subnet D.

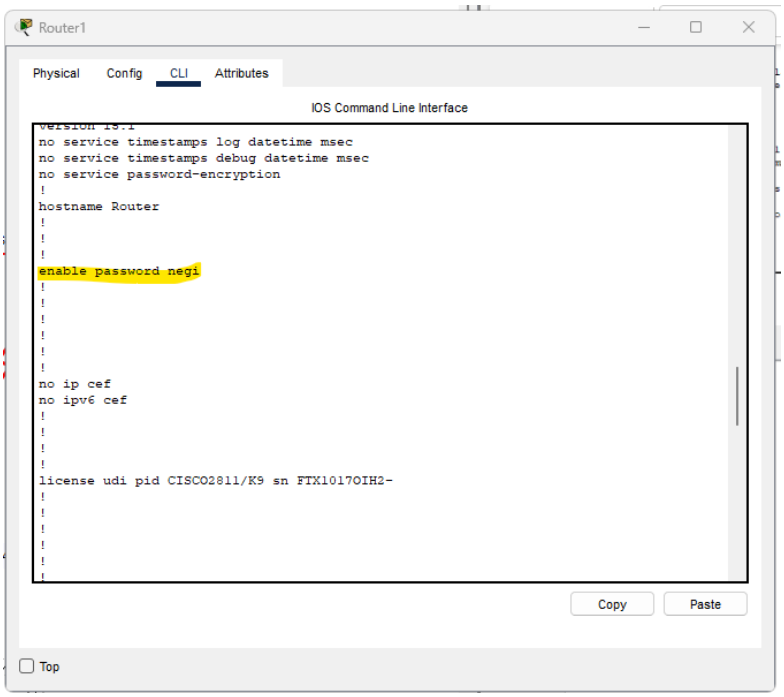


Figure 4

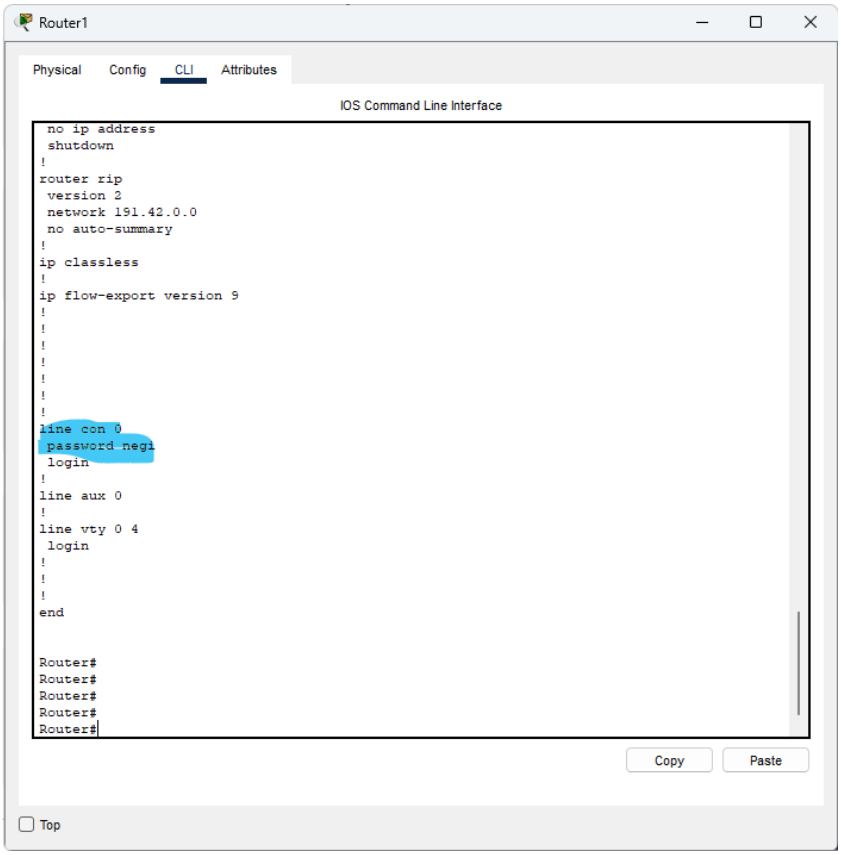

Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
no ip address
shutdown
!
router rip
version 2
network 191.42.0.0
no auto-summary
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
password negi
login
!
line aux 0
!
line vty 0 4
login
!
!
!
end
Router#
Router#
Router#
Router#
Router#
Copy
Paste
Top

Figure 5

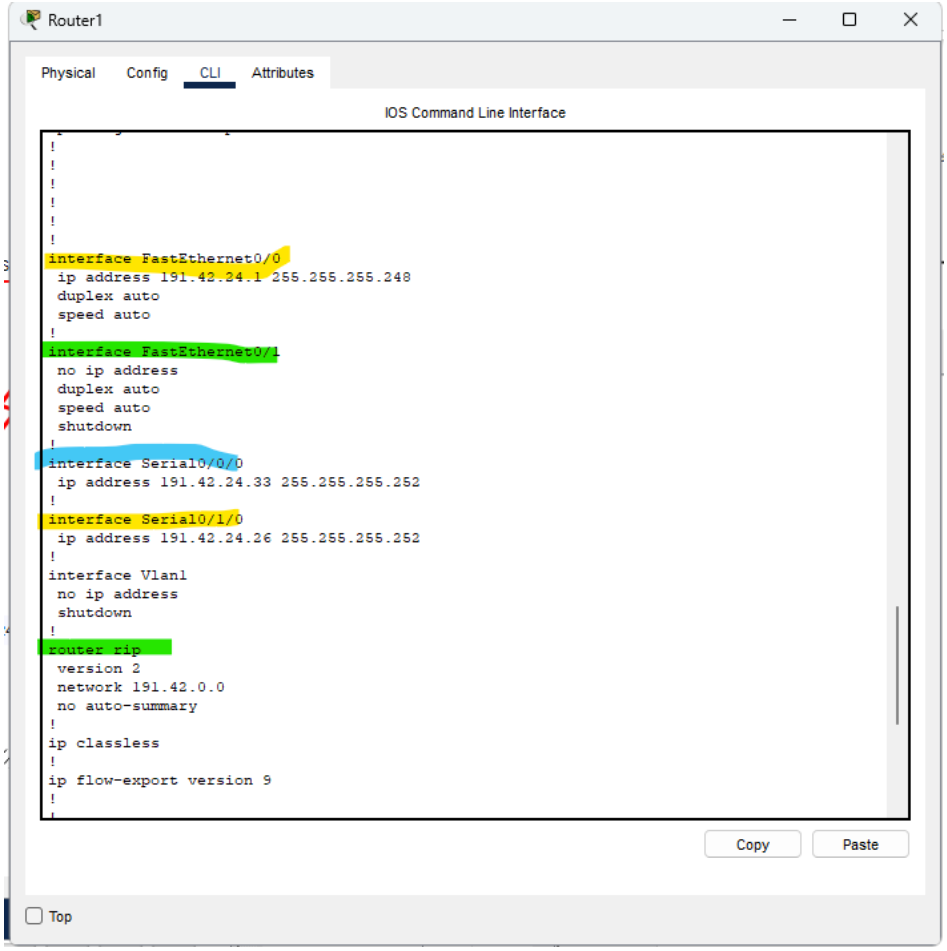


Figure 6

ROUTER 2

Router 2 for this router first there is enable password which is negi. The I configured Fast Ethernet 0/0 Ip 191.42.24.16 subnet 255.255.255.248. For configuring router rip there Ip is 191.42.0.0 and the serial0/0/0 Ip address 191.42.24.34 subnet 255.255.255.252 and serial 0/1/0 191.42.24.30 subnet 255.255.255.252 is configured with subnet D. Then Dhcp dynamic host control protocol for the connected devices.

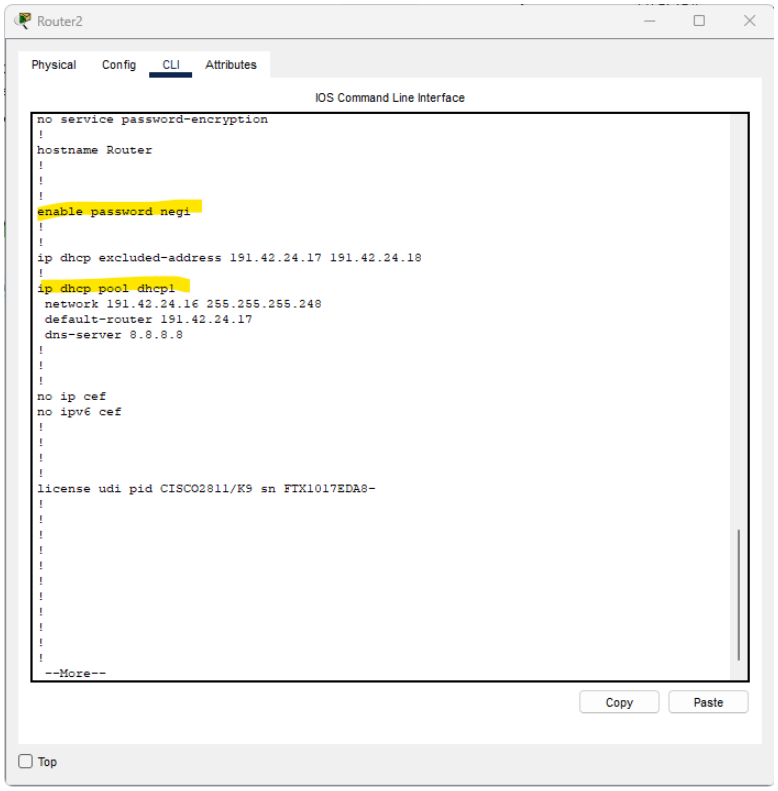

Router2
Physical
Config
CLI
Attributes
IOS Command Line Interface
no service password-encryption
!
hostname Router
!
!
!
enable password negi
!
!
ip dhcp excluded-address 191.42.24.17 191.42.24.18
!
ip dhcp pool dhcp1
network 191.42.24.16 255.255.255.248
default-router 191.42.24.17
dns-server 8.8.8.8
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2811/K9 sn FTX1017EDA8-
!
!
!
!
!
!
!
!
!
!
--More--
Copy
Paste
Top

Figure 7

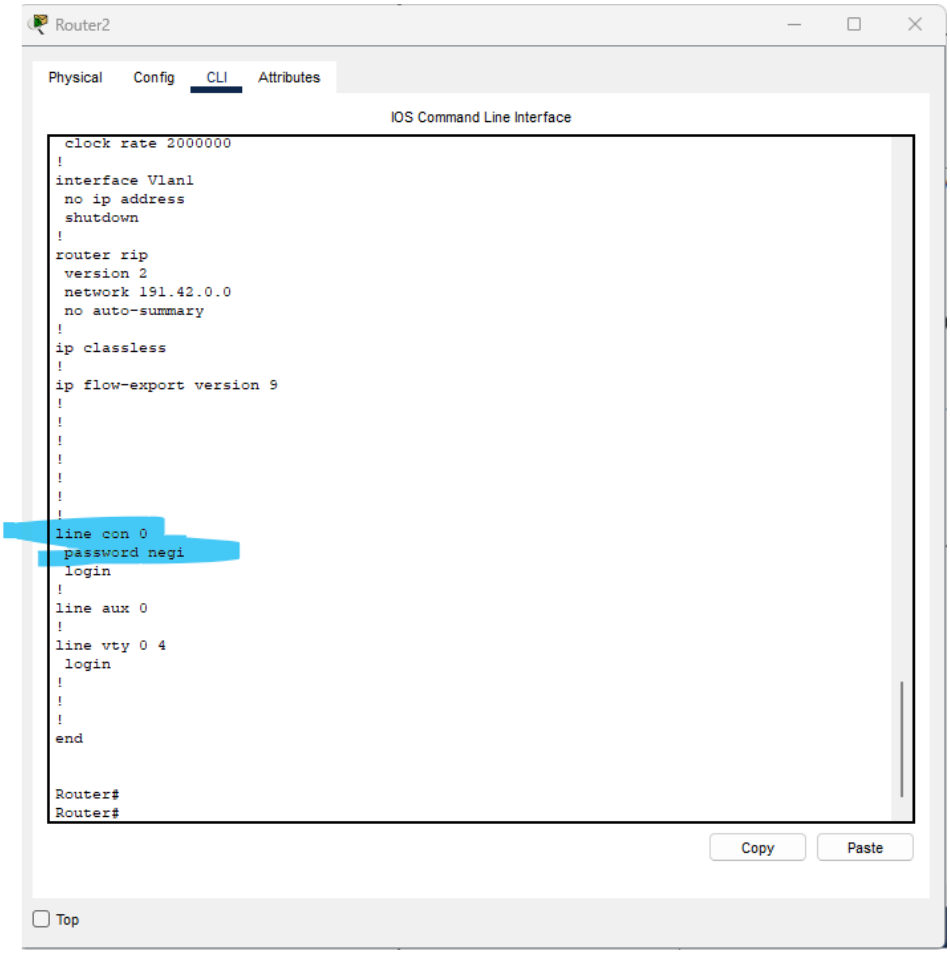

Router2
Physical
Config
CLI
Attributes
IOS Command Line Interface
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
network 191.42.0.0
no auto-summary
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
password negi
login
!
line aux 0
!
line vty 0 4
login
!
!
!
end
Router#
Router#
Copy
Paste
Top

Figure 9

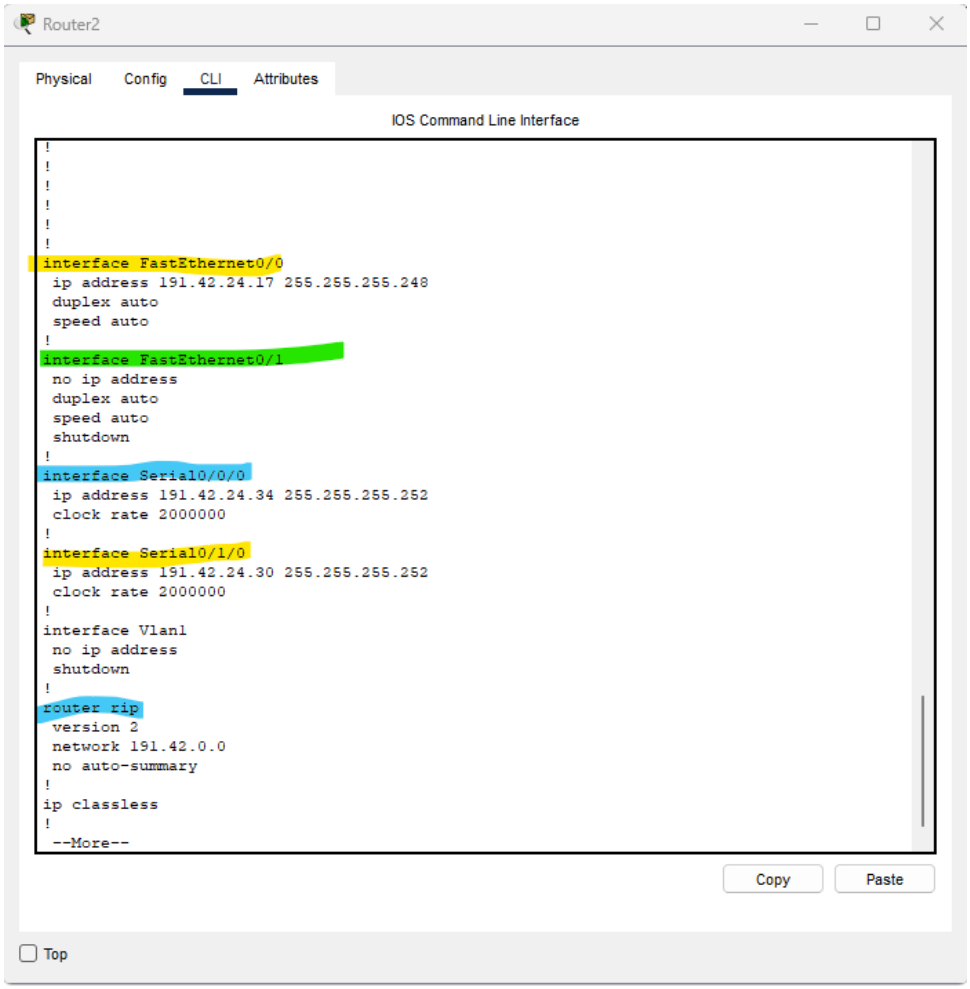


Figure 10

For wireless router0: configuring the Ip address 191.42.24.18 and subnet mask 255.255.255.248. DHCP pool is working from router 2. In wireless router DHCP is disabled

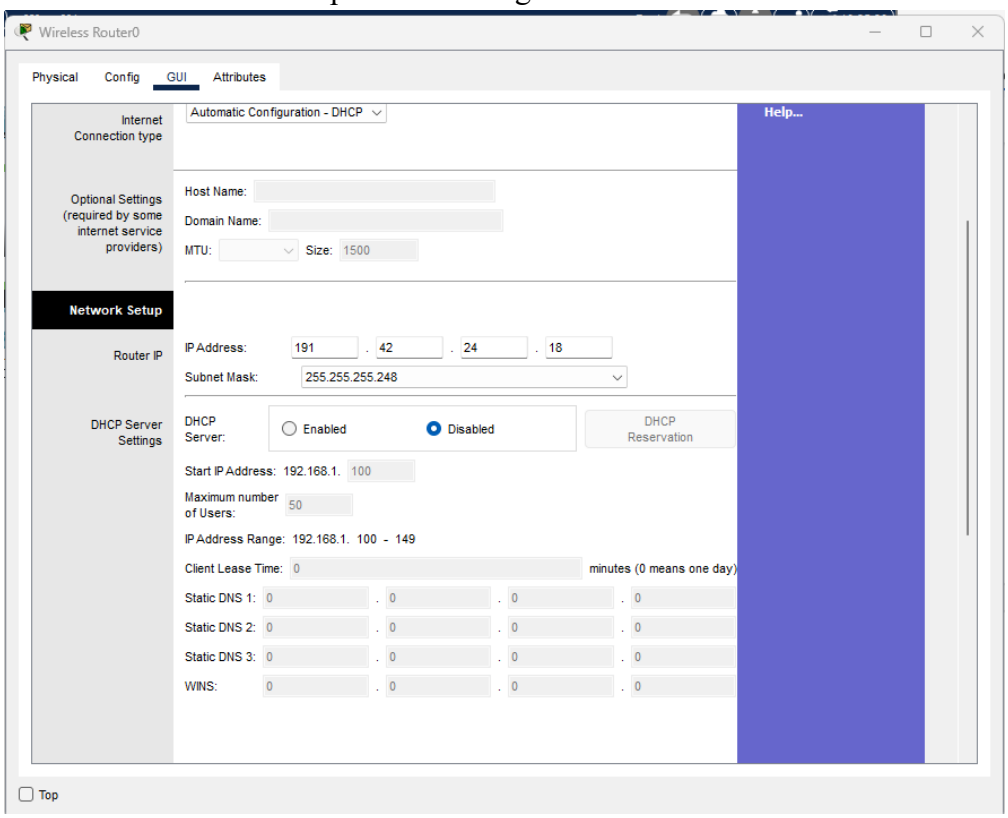


Figure 11

And for all pcs static Ip is given pc0 and server. But for wireless it is DHCP.

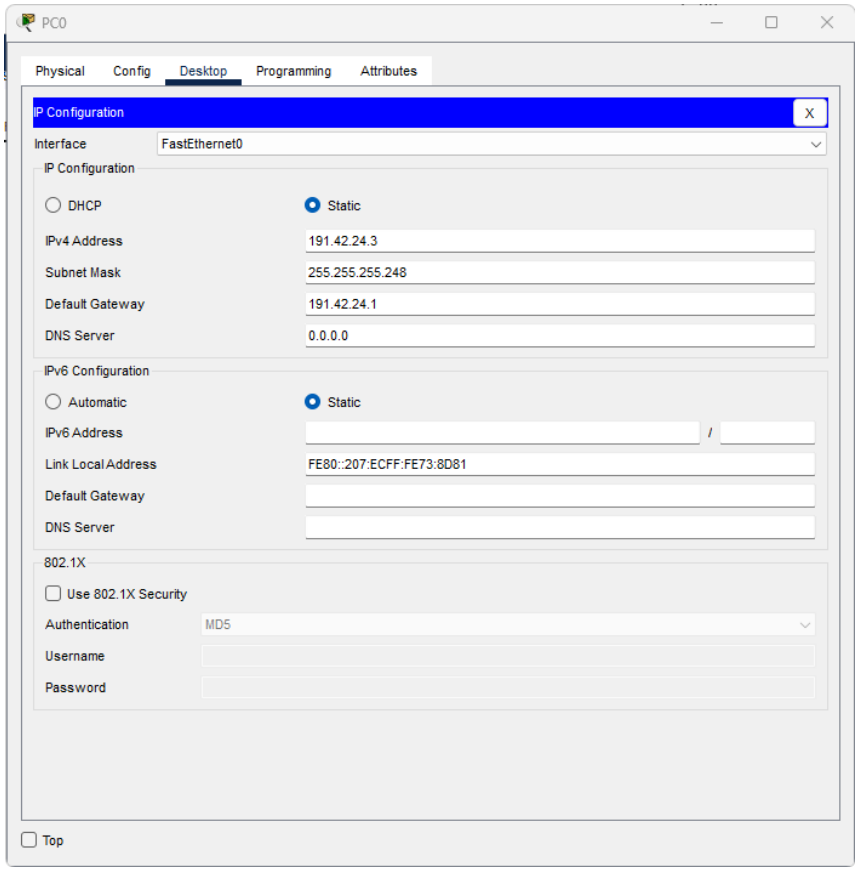


Figure 12

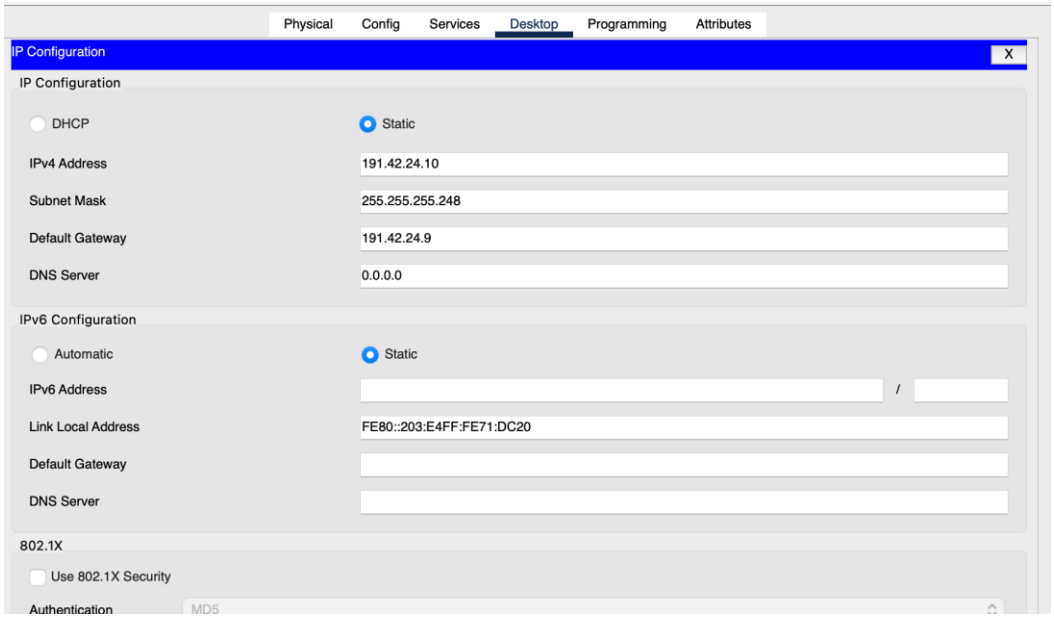


Figure 13

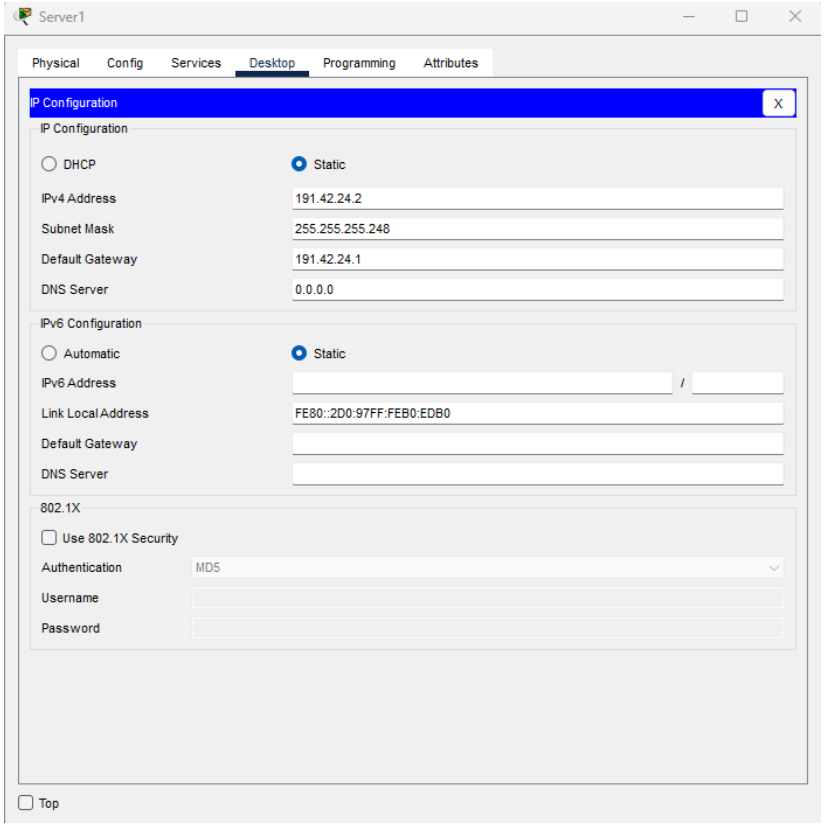

Server1
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
191.42.24.2
Subnet Mask
255.255.255.248
Default Gateway
191.42.24.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2D0:97FF:FEB0:EDB0
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

Figure 14

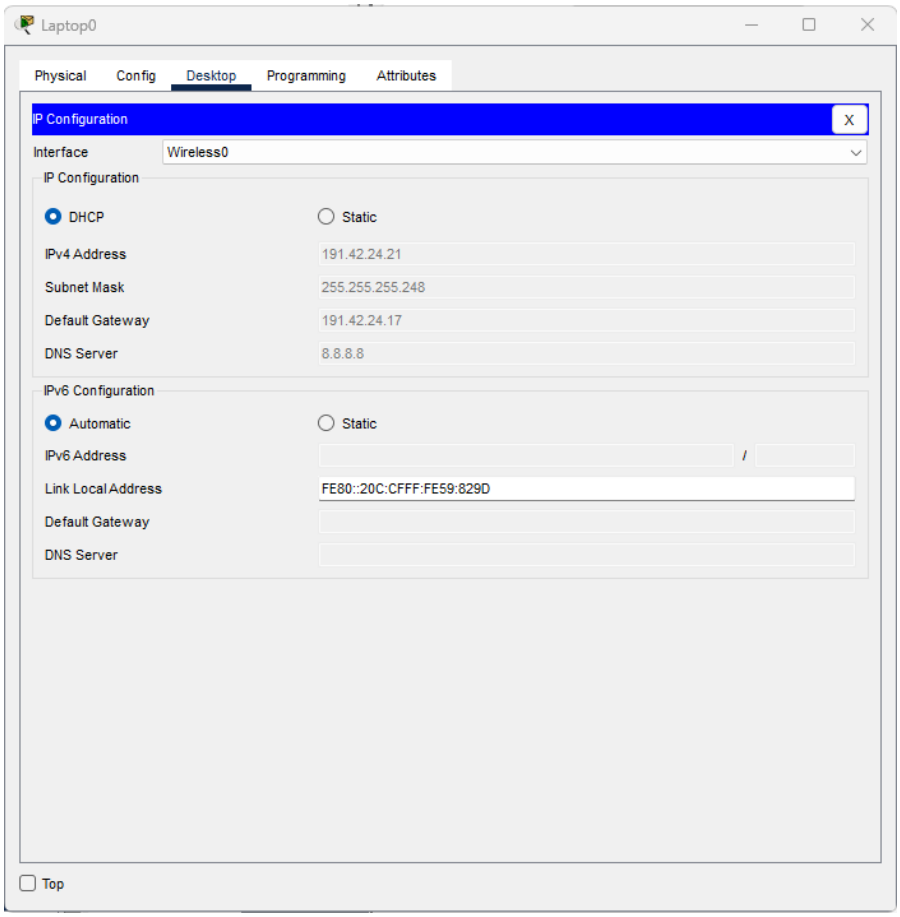


Figure 15

NETWORK DESIGN

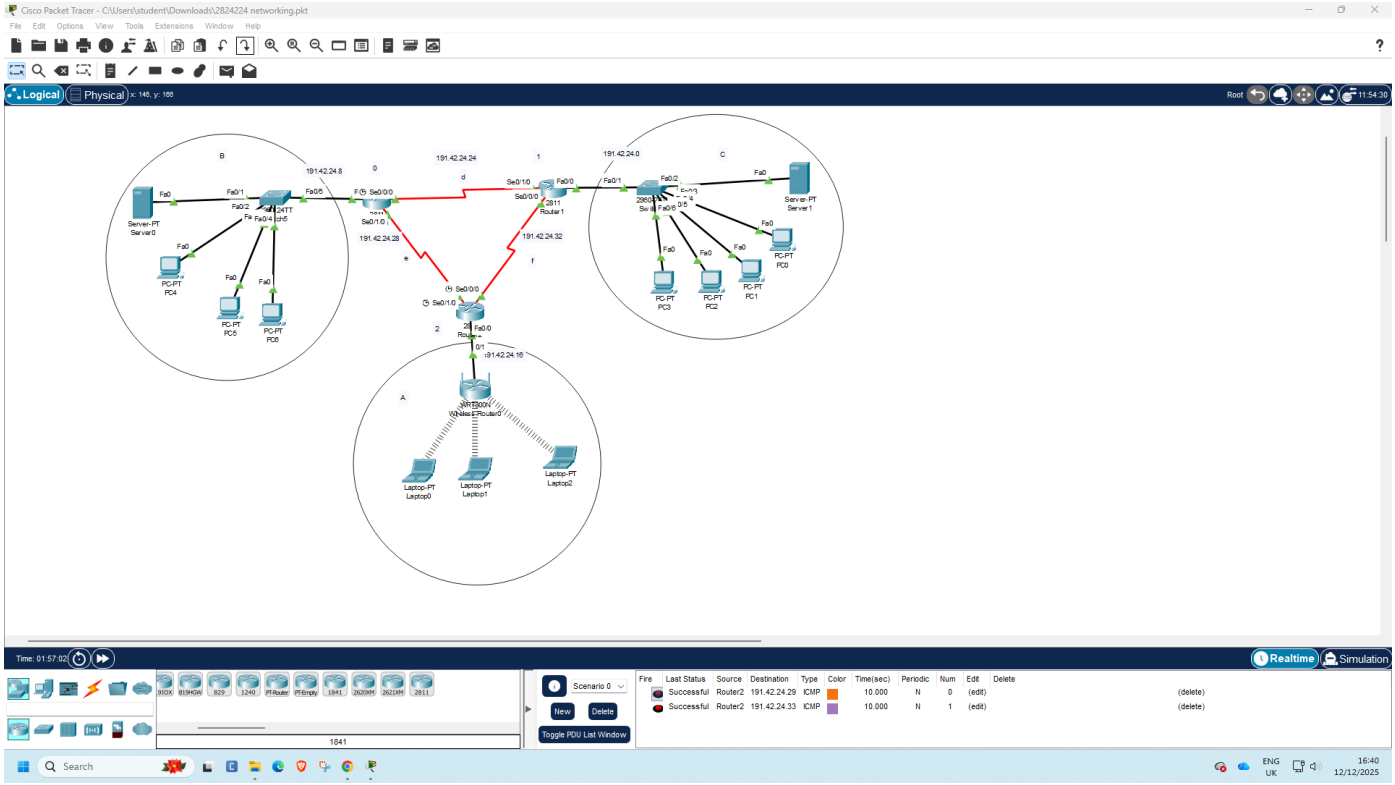


Figure 16

RESULT AND ANALYSIS (NETWORK VERIFICATION)

The whole network is completely configured, and it is vital to test it to check connections, so data is transferring to the whole devices connected to one another.

Here using PDU for some port which are mentioned. For other ports which are not specified there is simple PDU. Some screenshot to the right shows the same for Router0 Se0/0/0 to Router0 Fa 0/0 and the screenshot for the same router0 Se0/0/0 to Router1 Se0/0/0.

Router 0 serial 0/0/0 to router 1 serial 0/1/0, router 0 serial 0/1/0 to router2 serial 0/1/0 and router 0 to 191.42.24.9

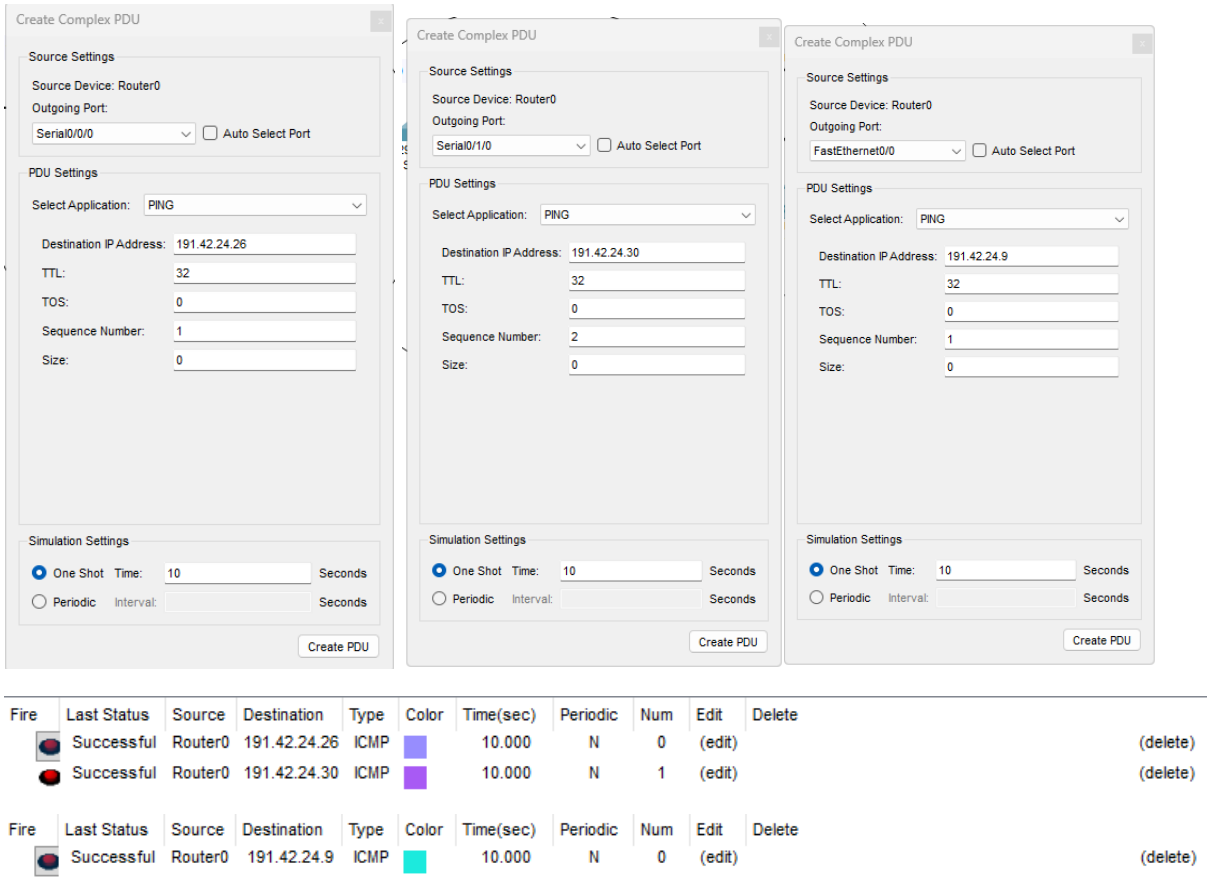
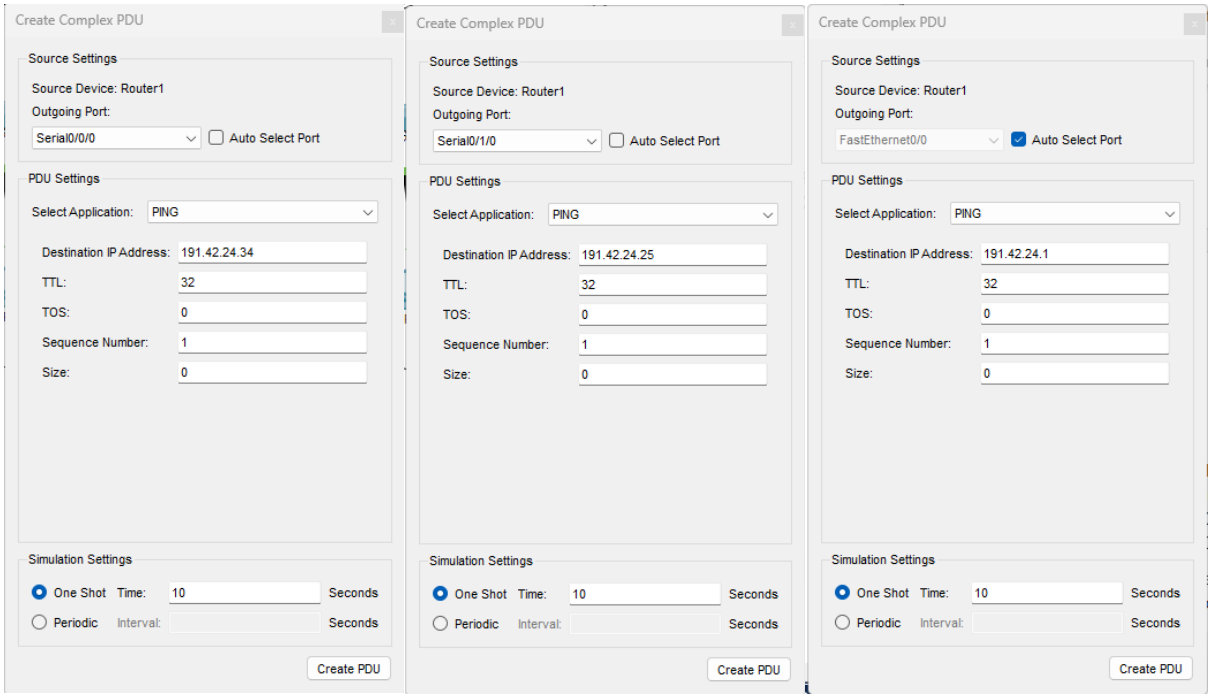
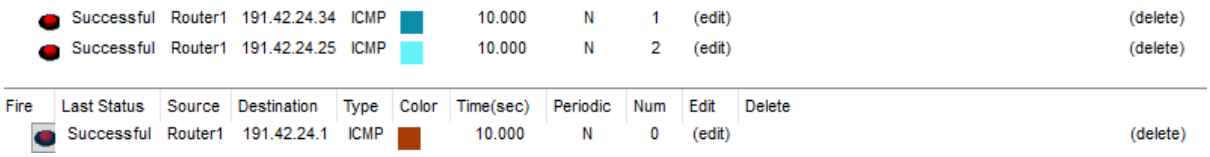


Figure 17

Router 1 serial 0/0/0 to router 2 serial 0/0/0, router 1 serial 0/1/0 to router 0 serial 0/1/0 and router 1 Fa 0/0 to 191.42.24.1





Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Router1	191.42.24.34	ICMP		10.000	N	1	(edit)	(delete)
	Successful	Router1	191.42.24.25	ICMP		10.000	N	2	(edit)	(delete)
	Successful	Router1	191.42.24.1	ICMP		10.000	N	0	(edit)	(delete)

Figure 18

Router 2 serial 0/0/0 to router 0 serial 0/1/0, router 2 serial 0/0/0 to router 1 serial 0/0/0 and router 2 to 191.42.24.17

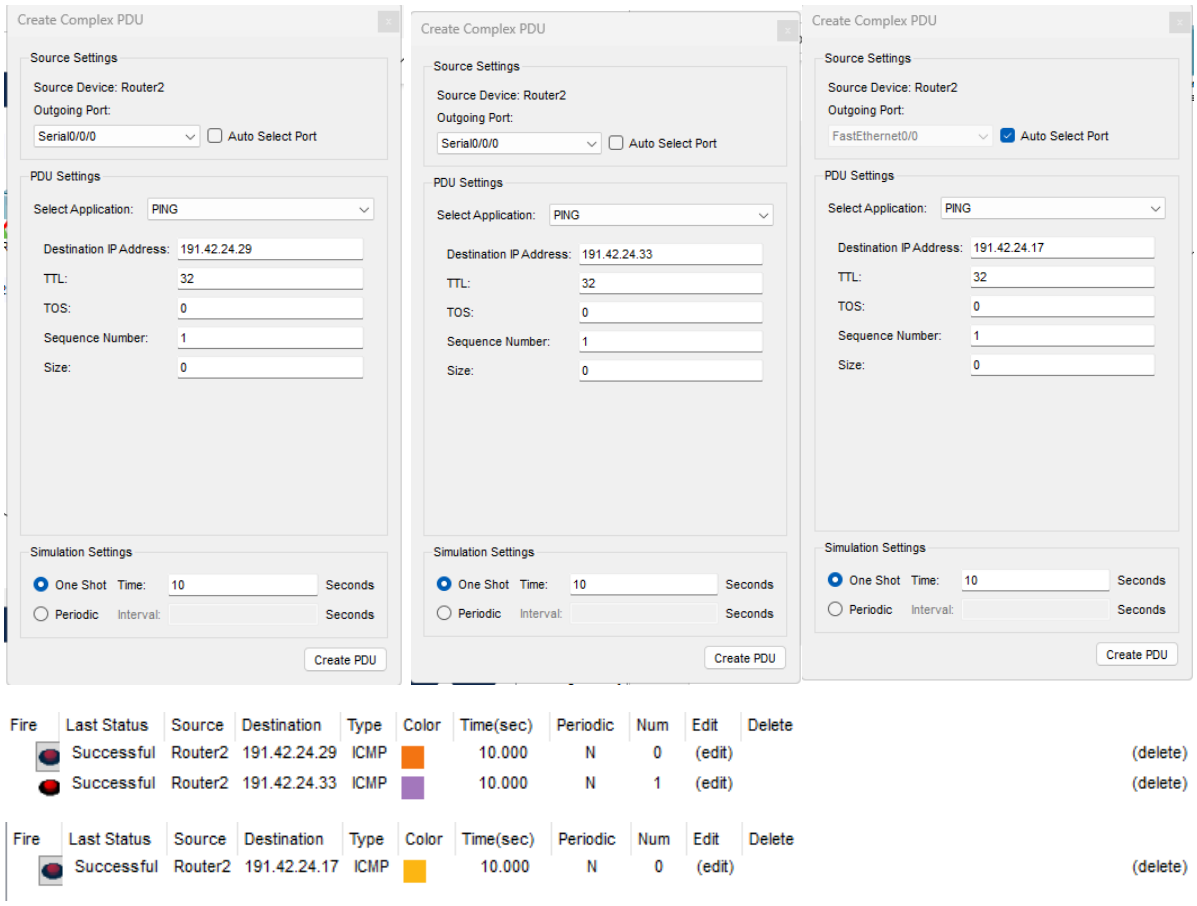


Figure 19

TASK 6 PING AND PDU

The table depicts the test results for the most important communications for each subnet. Networking on all over the network has been tested via PING and PDU requests displayed below.

SUBNET D

Table 8

From	To	IP Address (Destination)	Subnet (Destination)	Results(S/F)
Router 0, Se0/0/0	Router 0, Fa 0/0	191.42.24.9	b	S

	Router 0, Se 0/1/0	191.42.24.29	e	S
	Router1, Se0/0/0	191.42.24.33	f	S
	Router 1, Se0/1/0	191.42.24.26	d	S
	Router 1 Fa 0/0	191.42.24.1	c	S
	Router 2 serial 0/0/0	191.42.24.34	f	S
	Router 2 serial 0/1/0	191.42.24. 30	e	S
	Router 2 Fa0/0	191.42.24.17	a	S
Router 1 serial 0/1/0	Router 1, Fa 0/0	191.42.24.1	c	S
	Router 1, Se 0/0/0	191.42.24.33	f	
	Router 0, Se0/0/0	191.42.24.25	d	S
	Router 0, Se0/1/0	191.42.24.29	e	S
	Router 0, Fa 0/0	191.42.24.9	b	S
	Router 2 serial 0/0/0	191.42.24.34	f	S
	Router 2 serial 0/1/0	191.42.24.30	e	S
	Router 2 Fa0/0	191.42.24.17	a	S

SUBNET F

Table 8.1

From	To	IP Address (Destination)	Subnet (Destination)	Results(S/F)
Router 1, Se0/0/0	Router 1, Fa 0/0	191.42.24.1	c	S
	Router 1, Se0/1/0	191.42.24.26	d	
	Router 0, Se0/0/0	191.42.24.25	d	S
	Router 0, Se0/1/0	191.42.24.29	e	S
	Router 0 Fa 0/0	191.42.24.9	b	S
	Router 2 serial 0/0/0	191.42.24.34	f	S

	Router 2 serial 0/1/0	191.42.24.30	e	S
	Router 2 Fa0/0	191.42.24.17	a	S
Router 2, Se 0/0/0	Router 2, Fa 0/0	191.42.24.17	a	S
	Router 2, Se0/1/0	191.42.24.30	e	S
	Router 1, Se0/0/0	191.42.24.33	f	S
	Router1, Se0/1/0	191.42.24.26	d	S
	Router 1, Fa 0/0	191.42.24.1	c	S
	Router 0 Fa 0/0	191.42.24.9	b	S
	Router 0 serial 0/0/0	191.42.24.25	d	S
	Router 0 serial 0/1/0	191.42.24.29	e	S

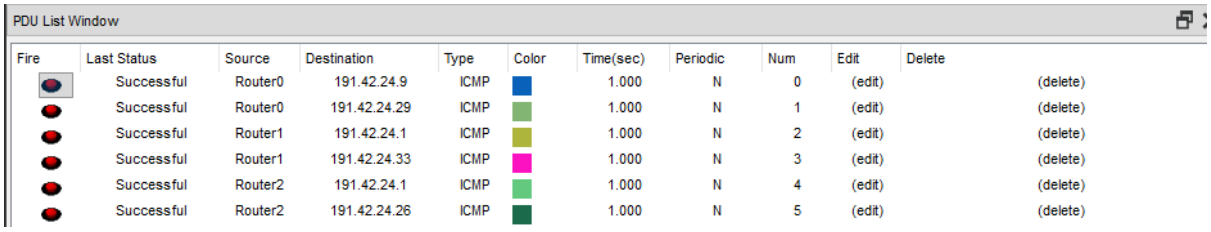
SUBNET E

Table 8.2

From	To	IP Address (Destination)	Subnet (Destination)	Results(S/F)
Router 0, Se0/0/0	Router 0, Fa 0/0	191.42.24.9	b	S
	Router 0, Se0/1/0	191.42.24.29	e	S
	Router 1, Se0/0/0	191.42.24.33	f	S
	Router 1, Se0/1/0	191.42.24.26	d	S
	Router 1 Fa 0/0	191.42.24.1	c	S
	Router 2 serial 0/0/0	191.42.24.34	f	S
	Router 2 serial 0/1/0	191.42.24.30	e	S
	Router 2 Fa0/0	191.42.24.17	a	S
Router 1, Se0/1/0	Router 1, Fa 0/0	191.42.24.1	c	S
	Router 1, Se 0/0/0	191.42.24.33	f	S

	Router 0, Se0/0/0	191.42.24.25	d	S
	Router 0, Se0/1/0	191.42.24.29	e	S
	Router 0 Fa 0/0	191.42.24.25	b	S
	Router 2 serial 0/0/0	191.42.24.34	f	S
	Router 2 serial 0/1/0	191.42.24. 30	e	S
	Router 2 Fa 0/0	191.42.24.17	a	S

List of the PDU depicted by the screenshot.



PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Router0	191.42.24.9	ICMP		1.000	N	0	(edit)	(delete)
	Successful	Router0	191.42.24.29	ICMP		1.000	N	1	(edit)	(delete)
	Successful	Router1	191.42.24.1	ICMP		1.000	N	2	(edit)	(delete)
	Successful	Router1	191.42.24.33	ICMP		1.000	N	3	(edit)	(delete)
	Successful	Router2	191.42.24.1	ICMP		1.000	N	4	(edit)	(delete)
	Successful	Router2	191.42.24.26	ICMP		1.000	N	5	(edit)	(delete)

FIGURE 20

PINGING DEVICES TO CHECK THE NETWORK

Here, for the devices in subnet c, b and a

Subnet c to subnet a, d, e, f, and b.

Table 9

From	To	Ip address	Result (S/F)
Server 1	Pc 0	191.42.24.3	S
	Router 1	191.42.24.1	S
	Router 0	191.42.24.9	S
	Server 0	191.42.24.10	S
	Router 2	191.42.24.17	S
	Laptop0	191.42.24.19	S

Subnet b to subnet a, c, d, e, and f.

Table 9.1

From	To	Ip address	Result (S/F)
Server 0	Pc4	191.42.24.11	S
	Router 0	191.42.24.9	S
	Router 1	191.42.24.1	S
	Server 1	191.42.24.2	S

	Router 2	191.42.24.17	S
	Laptop0	191.42.24.19	S

Subnet a to subnet c, b, d, e and f.

Table 9.2

From	To	Ip address	Result (S/F)
Laptop0	Laptop 1	191.42.24.20	S
	Router 2	191.42.24.17	S
	Router 0	191.42.24.9	S
	Server 0	191.42.24.10	S
	Router 1	191.42.24.1	S
	Server1	191.42.24.2	S

SCREENSHOT OF PING OF THE NETWORK TO DISPLAY THAT THE NETWORK IS WORKING PERFECTLY.

Ping testing of Subnet c

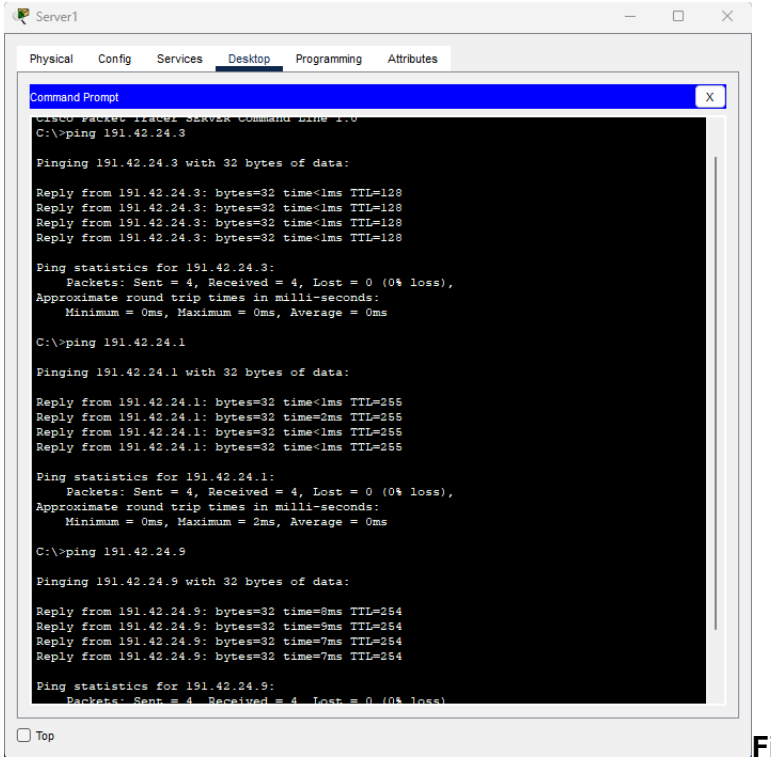


Figure 21

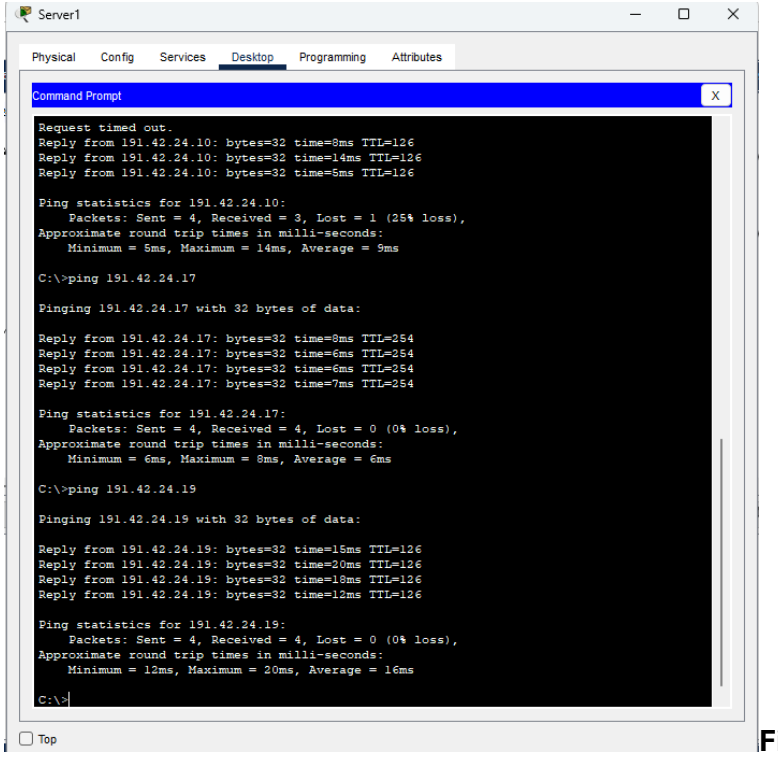


Figure22

Screenshot of pinging of Subnet b

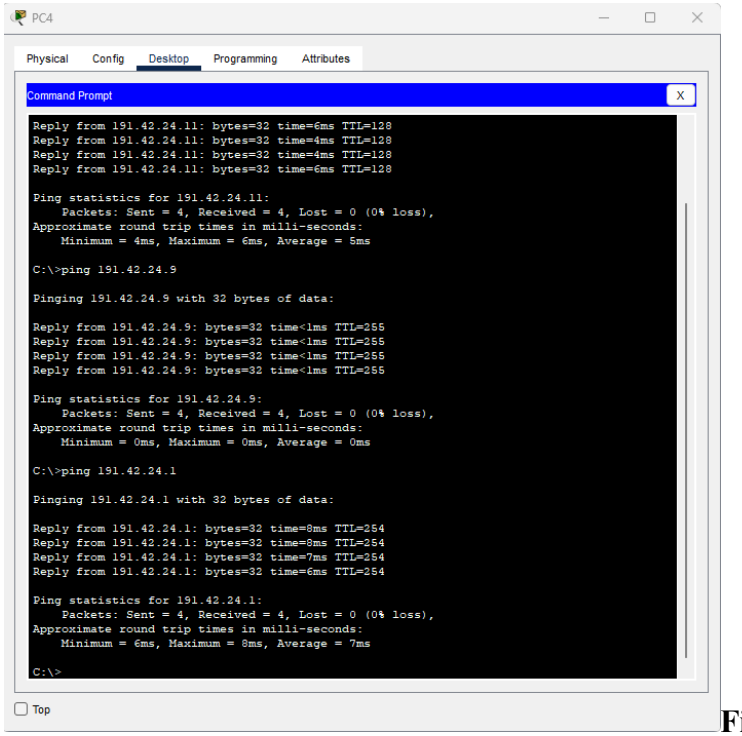


Figure 23

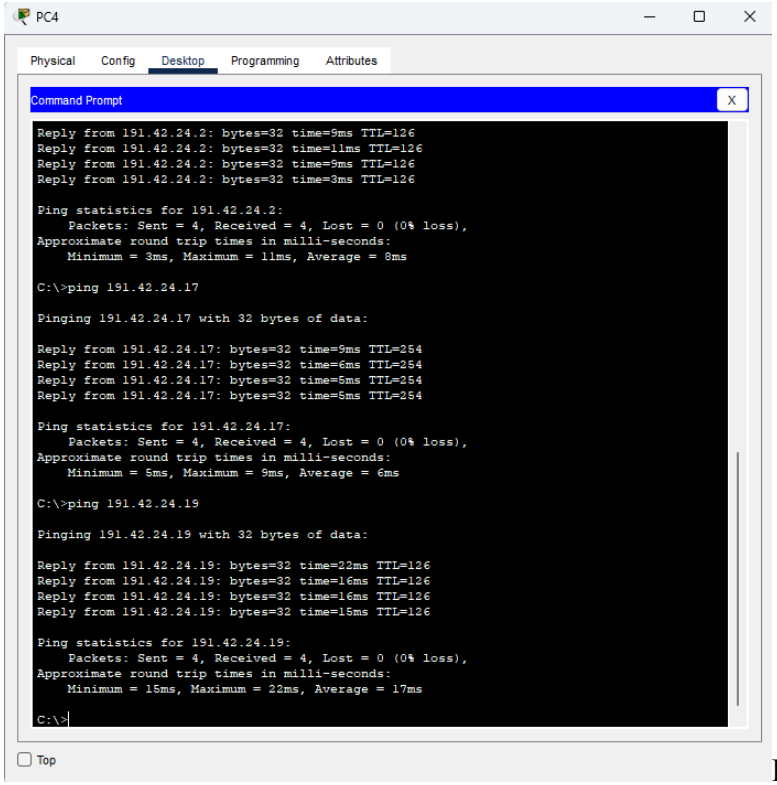


Figure 24

Screenshot of subnet a

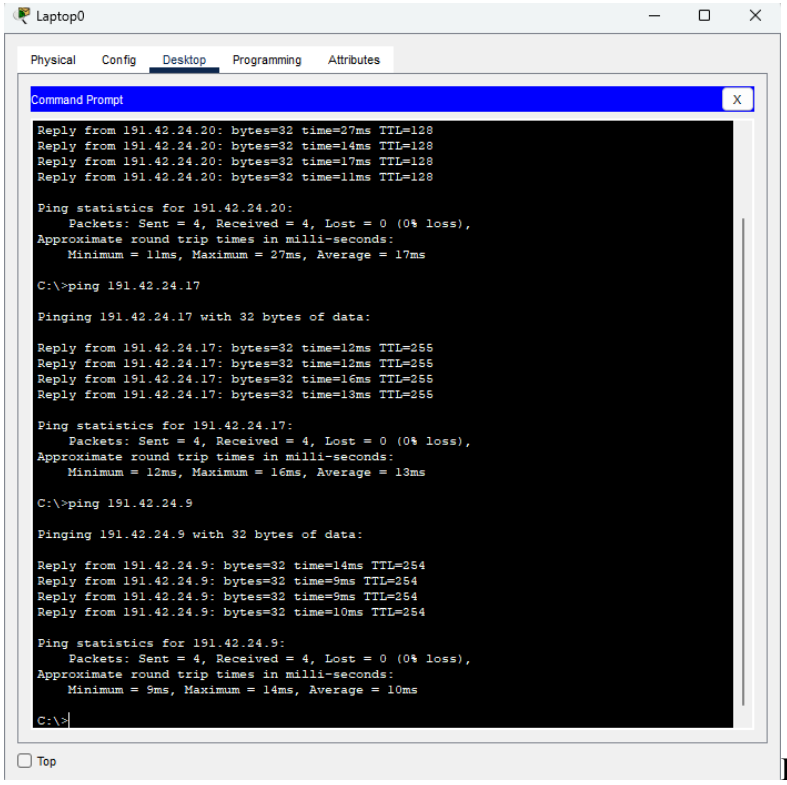


Figure 25

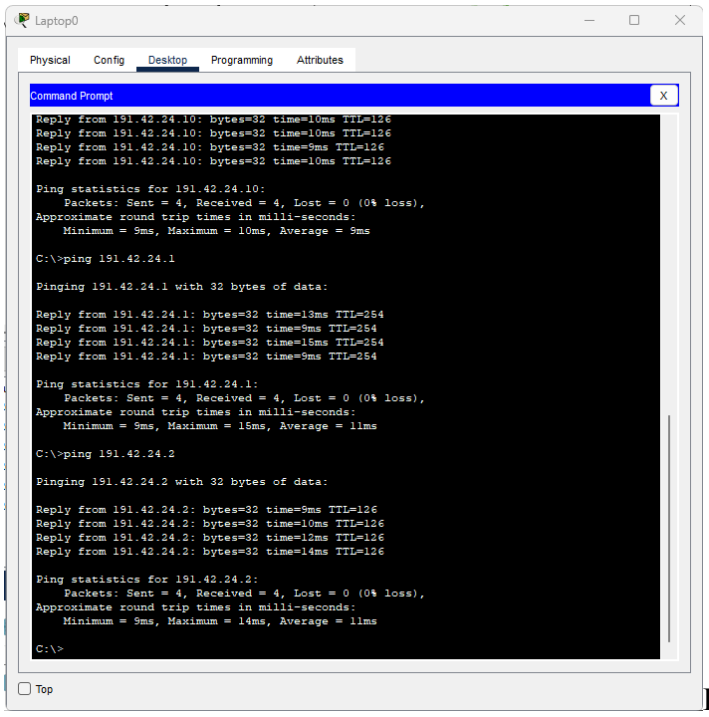


Figure 26

EVALUATION

The tables above elaborate the configuration of the network. This working also demonstrates that the IPv4 and VLSM calculations are correct for the required network configuration. The RIP protocols are working well and configured in my network. During the initial stages of run there is problem in router 2 which is connected to wireless devices i.e. laptops via wireless router i.e. fixed later while using DHCP pool in router 2.

For detailed explanation tables and figures, are more suitable for the better understanding.

ARP

In the cisco packet tracer during network simulation, in the initial stage the packet sent form a host device in one subnet to a host device on another subnet results failed due to the routers were taking time to boot. After the first attempt failed, retrying to send the packet again and then it will be successful. The success is only cause of the ARP table was successfully filled by providing the information on both ends as well as devices connected between them.

EVALUATION OF SIMULATION

The network simulation all the devices are configured and assigned the correct commands for the implementation on the Cisco packet tracer. The hosts/servers are configured by static IP meanwhile the devices in subnet are configured via DHCP pool of Ip address. The whole network is tested with PDU and ping commands. The simulation makes it possible for network to communicate with each other. There all the test performed results successful.

All the DHCP, RIPv2 and statically assigned Ip of hosts/servers are shown by screenshots and tables when the PDU and ping were performed where they all succeed in sending and receiving packets.

CHALLENGES

During the testing of network- Overall network works fine in the initial stages of testing except the router 2 it was not taking the Rip commands for the communication of files. Because there was error in config of RIP addresses after assigned them in order it started working and the whole network is communicating and running fine. Also, the wireless was not working correctly while assigning IPs to the devices connected to it. So, we directly configured router 2 for the DHCP pool for that and then it began to talk with other devices.

Conclusion

Conclusively, the network consisted of 18 devices in the network. These devices included 7 pcs, 3 laptops, 2 servers and 1 access point. The necessary connecting devices were mentioned in above TABLE for the clear specifications of the devices used between them. The Subnet D, E and F consisted of 2 routers with no hosts.

Subnet C had most devices which includes 1 server ,4 pcs and 1 switch to connect with router and all the devices IP configured statically. Subnet A had 4 devices including 1 access point and 3 laptops there IP was assigned by DHCP pool from connected router and the wireless router is secured by WPA2-PSK for the speed and security purpose. At last subnet B which has 4 devices includes 1 server, 3 pcs and 1 switch to connect with router and they all configured statistically as well. RIPv2 is used for connecting the networks with each other using CLI commands. Meanwhile all the subnets were calculated with VLSM depicted in above table. With all this the network derived successfully in communicating with each other.

REFERENCES

- Cisco Networking Academy (2023) *Introduction to Networks v7.0*. Cisco Systems Inc. Available at: https://www.netacad.com (Accessed: 10 December 2025).
- Cox, J., 2020. Introduction to Subnetting – Calculate Subnets, CIDR, VLSM.
- Tomsho, G. (2011) *Guide to Networking Essentials*. 6th edn. Boston: Course Technology, Cengage Learning.
- Kurose, J.F. and Ross, K.W. (2023) *Computer Networking: A Top-Down Approach*. 8th edn. Boston: Pearson.
- Comer, D.E. (2021) *Internetworking with TCP/IP, Volume 1: Principles, Protocols, and Architecture*. 6th edn. Boston: Pearson Education.